

CS 255 Business Requirements Document

System Components and Design

Purpose

- DriverPass wants to provide customers with online driver's education through practice tests, online classes, and behind-the-wheel driving lessons to better prepare them for the DMV driving exam.
- The purpose of this project is to design a cloud-based system that allows customers to register, purchase training packages, schedule driving lessons, complete online practice tests, and access learning materials from any computer or mobile device with an internet connection.
- The system will allow DriverPass employees, instructors, secretaries, IT administrators, and the company owner to manage customers, appointments, reports, and user accounts according to their assigned roles.
- The system should improve scheduling efficiency, reduce manual record keeping, and provide customers with convenient online access while helping DriverPass manage its daily operations more effectively.

System Background

- Many people fail their DMV driving test because they lack adequate preparation. Liam created DriverPass to provide customers with better training opportunities through online learning and driving instruction.
- DriverPass currently needs a centralized system that can manage customer information, appointments, online learning, driving instructors, vehicles, and reports.
- Customers should be able to register for training packages, schedule driving lessons online, complete practice tests, and review their progress through the system.
- Employees should be able to manage reservations, customer records, schedules, and reports while administrators maintain security and user accounts.
- The system must also remain current with DMV rule changes by receiving updated policies, rules, and sample questions from the DMV whenever changes occur.

Objectives and Goals

- Provide customers with an online account that allows them to manage their personal information and access DriverPass services.
- Allow customers to schedule, cancel, and modify two-hour driving lesson appointments online.
- Allow the secretary to schedule and manage appointments for customers who call or visit the DriverPass office.
- Track each driving appointment by customer, driver, vehicle, date, and time.

- Allow DriverPass to manage its three driving packages and disable a package when it is no longer available for registration.
- Provide customers with access to online classes and practice tests when included in their selected package.
- Track online test progress, including the test name, time taken, score, and status.
- Allow authorized employees to view driver notes and lesson information.
- Provide role-based access so employees receive only the system access appropriate to their responsibilities.
- Maintain an activity history showing who created, canceled, or modified reservations so DriverPass can identify who made a change if a problem occurs.
- Allow DriverPass to receive notifications when the DMV updates rules, policies, or sample questions.
- Allow authorized users to access system information from computers and mobile devices.
- Allow authorized users to download reports and information that can be used in programs such as Microsoft Excel.

Requirements

Nonfunctional Requirements

- The system shall operate as a web-based application and preferably be hosted through a cloud environment.
- The system shall allow authorized users to access DriverPass information from computers and mobile devices with an internet connection.
- The system shall allow customers to access their accounts, view available driving appointments, schedule appointments, cancel appointments, and modify appointments online.
- The system shall allow authorized users to access business information without requiring DriverPass employees to maintain local servers.
- The system shall provide timely access to customer, appointment, driver, vehicle, package, and test information during normal business operations.
- The system should minimize technical problems that could interfere with DriverPass's ability to operate the business.
- The system shall allow authorized users to download reports and information for use outside the system, including in spreadsheet software such as Microsoft Excel.

Performance Requirements

- The system shall operate through a web-based interface rather than requiring customers to install a dedicated desktop application.
- The system should be hosted in a cloud environment, so DriverPass does not have to manage its own server backups and security infrastructure.
- The system shall support access from computers and mobile devices.

- The system shall use a centralized database to store customer, appointment, driver, vehicle, package, user, and test information.
- The system shall allow authorized users to access the same current information from supported devices.
- The system shall support exporting reports and selected information for use in spreadsheet software such as Microsoft Excel.

Platform Constraints

- The system shall operate through a web-based interface rather than requiring customers to install a dedicated desktop application.
- The system should be hosted in a cloud environment, so DriverPass does not have to manage its own server backups and security infrastructure.
- The system shall support access from computers and mobile devices.
- The system shall use a centralized database to store customer, appointment, driver, vehicle, package, user, and test information.
- The system shall allow authorized users to access the same current information from supported devices.
- The system shall support exporting reports and selected information for use in spreadsheet software such as Microsoft Excel.

Accuracy and Precision

- The system shall assign each user an account and role so customers, secretaries, IT staff, and other employees can be distinguished from one another.
- The system shall associate each driving reservation with the correct customer, driver, vehicle, date, and time.
- The system shall prevent conflicting reservations from assigning the same driver or vehicle to multiple customers for the same time period.
- The system shall record the start and end times for each driving lesson.
- The system shall maintain accurate customer information, including first name, last name, address, state, phone number, and other required registration information.
- The system shall maintain accurate package information, so customers receive the services associated with the package they selected.
- The system shall accurately track online test information, including test name, time taken, score, and status.
- The system shall identify test status as not taken, in progress, failed, or passed.
- The system shall record which user created, canceled, or last modified a reservation.
- The system shall notify appropriate users when information required for a transaction is missing or invalid.

Adaptability

- The system shall allow authorized administrators to add, modify, disable, and manage user accounts without changing the application code.
- The IT officer shall have sufficient administrative access to maintain the system and manage user accounts.
- The IT officer shall be able to reset a user's password when the user forgets it.
- The IT officer shall be able to block a user's access when the user's employment ends or access is otherwise revoked.
- The system shall allow authorized users to disable a driving package so that new customers cannot register for the package.
- The system should be designed so that future system updates can accommodate additional packages or features.
- Adding or removing major system modules may require assistance from a developer or system analyst and is outside the required functionality of the initial release.

Security

- The system shall require users to authenticate before accessing information associated with their accounts.
- The system shall assign users different roles and permissions based on their responsibilities within DriverPass.
- The IT officer shall have full administrative access necessary to maintain accounts, reset passwords, and block user access.
- The system shall restrict customers from accessing administrative functions or other customers' information.
- The system shall allow users to automatically reset their passwords when they forget them.
- The system shall allow the IT officer to disable a user's access when the employee is no longer authorized to use the system.
- The system shall maintain an activity history identifying the user who created, canceled, or modified a reservation.
- The system shall protect sensitive customer information, including payment information collected during registration.
- The system should use secure connections to protect information transmitted between users and the cloud-based system.

Functional Requirements

- The system shall allow customers to create an account and provide required registration information, including their first name, last name, address, state, phone number, payment information, pickup location, and drop-off location.

- The system shall allow customers to log in to their accounts and access services associated with their accounts.
- The system shall allow customers to schedule two-hour driving lessons by selecting an available day and time.
- The system shall allow customers to cancel driving lesson reservations through their accounts.
- The system shall allow customers to modify existing driving lesson reservations through their accounts.
- The system shall allow the secretary to create, modify, and cancel customer appointments when customers schedule lessons by phone or in person at the DriverPass office.
- The system shall associate each customer reservation with the assigned driver and vehicle.
- The system shall prevent scheduling conflicts involving the same driver, vehicle, date, and time.
- The system shall allow customers to select from the available DriverPass driving packages.
- The system shall allow authorized employees to disable a package so that customers cannot register for a package that is no longer available.
- The system shall provide the services associated with the customer's selected package, including driving lessons, in-person instruction, online classes, and practice tests when applicable.
- The system shall allow customers with access to online training to complete practice tests.
- The system shall track each customer's online test progress, including test name, time taken, score, and status.
- The system shall identify test status as not taken, in progress, failed, or passed.
- The system shall allow authorized users to view driver notes and lesson information, including lesson times and driver comments.
- The system shall allow authorized users to view customer information and update appropriate records based on their role.
- The system shall record who created, canceled, or last modified a reservation.
- The system shall allow authorized users to generate and print activity reports to determine who made changes to reservations.
- The system shall allow authorized users to download reports and information for use in spreadsheet software such as Microsoft Excel.
- The system shall allow the IT officer to reset passwords and block user accounts.
- The system shall allow users to automatically reset forgotten passwords.
- The system shall receive updates from the DMV regarding rules, policies, and sample questions and notify DriverPass when updates are available.
- The system shall provide a contact page that allows customers to contact DriverPass.
- The system shall provide a method for DriverPass employees to contact customers using information stored in the system.

User Interface

- The system shall provide a web-based interface that customers and employees can access from computers and mobile devices.
- The customer interface shall allow customers to view and manage their personal information.

- The customer interface shall allow customers to schedule, cancel, and modify driving lesson reservations.
- The customer interface shall allow customers to view their online test progress.
- The online test progress area shall display the test name, time taken, score, and status.
- The customer interface shall provide access to online classes and practice tests when included in the customer's selected package.
- The system shall provide an input form that allows the customer or secretary to enter and manage customer information.
- The system shall provide an area for driver notes that displays lesson times and driver comments.
- The system shall provide an area for special needs information, a driver photo, and a student photo as identified in the client's interface design.
- The system shall provide a contact page for communicating with DriverPass.
- The secretary interface shall allow the secretary to enter customer information and schedule appointments for customers who call or visit the office.
- The administrative interface shall provide authorized personnel with access to manage users, appointments, drivers, vehicles, packages, and system information according to their assigned roles.
- The interface shall use role-based access so users see functions appropriate to their responsibilities.

Assumptions

- The system assumes customers have access to a computer or mobile device and an internet connection because customers will use the web-based system to manage appointments, access online classes, and complete practice tests.
- The system assumes DriverPass employees have access to internet-connected devices needed to manage customer and business information.
- The system assumes DriverPass will provide accurate customer, driver, vehicle, and package information for entry into the system.
- The system assumes DriverPass will maintain accurate information about driver availability and vehicle availability so appointments can be scheduled correctly.
- The system assumes DriverPass will identify appropriate roles and permissions for its owner, IT officer, secretary, and other employees.
- The system assumes DriverPass will provide the information necessary to keep training materials and practice tests aligned with current DMV rules and requirements.
- The system assumes the DMV will provide DriverPass with updates regarding rules, policies, and sample questions through a connection or method that the system can use.
- The system assumes a cloud or hosting provider will be available to support the web-based system.
- The system assumes a third-party service may be required for certain functions, such as payment processing, and that the availability of such services may affect system functionality.

Limitations

- The system will depend on an internet connection for users to access current information and perform online functions such as scheduling appointments, accessing online classes, and completing practice tests.
- Although Liam wants to access data offline, the system will not allow users to modify or update system data while offline because doing so could result in duplicate or conflicting data.
- Users who do not have reliable internet access may have difficulty accessing online training, managing appointments, or completing practice tests.
- The system will depend on the availability of the cloud or hosting provider. A cloud service outage could temporarily prevent DriverPass employees and customers from accessing the system.
- The system may depend on third-party services for functions such as payment processing. If a third-party service experiences an outage or technical problem, related DriverPass functions may be unavailable even when the DriverPass application is operating.
- The system's connection with the DMV may depend on the availability and format of information provided by the DMV. Changes to DMV systems, rules, policies, or data formats could require additional development and testing.
- Major changes to the system, such as adding or removing system modules, may require assistance from a developer or system analyst and may need to be addressed in a future release.
- Changes to requirements after development has begun may require additional design, development, and testing time and could affect the project schedule.
- The initial system will focus on the requirements identified during the current project and may not include future features that DriverPass has not yet defined.
- Users who have difficulty using computers or mobile devices may require additional assistance or training to use the system.

Gantt Chart

DriverPass

